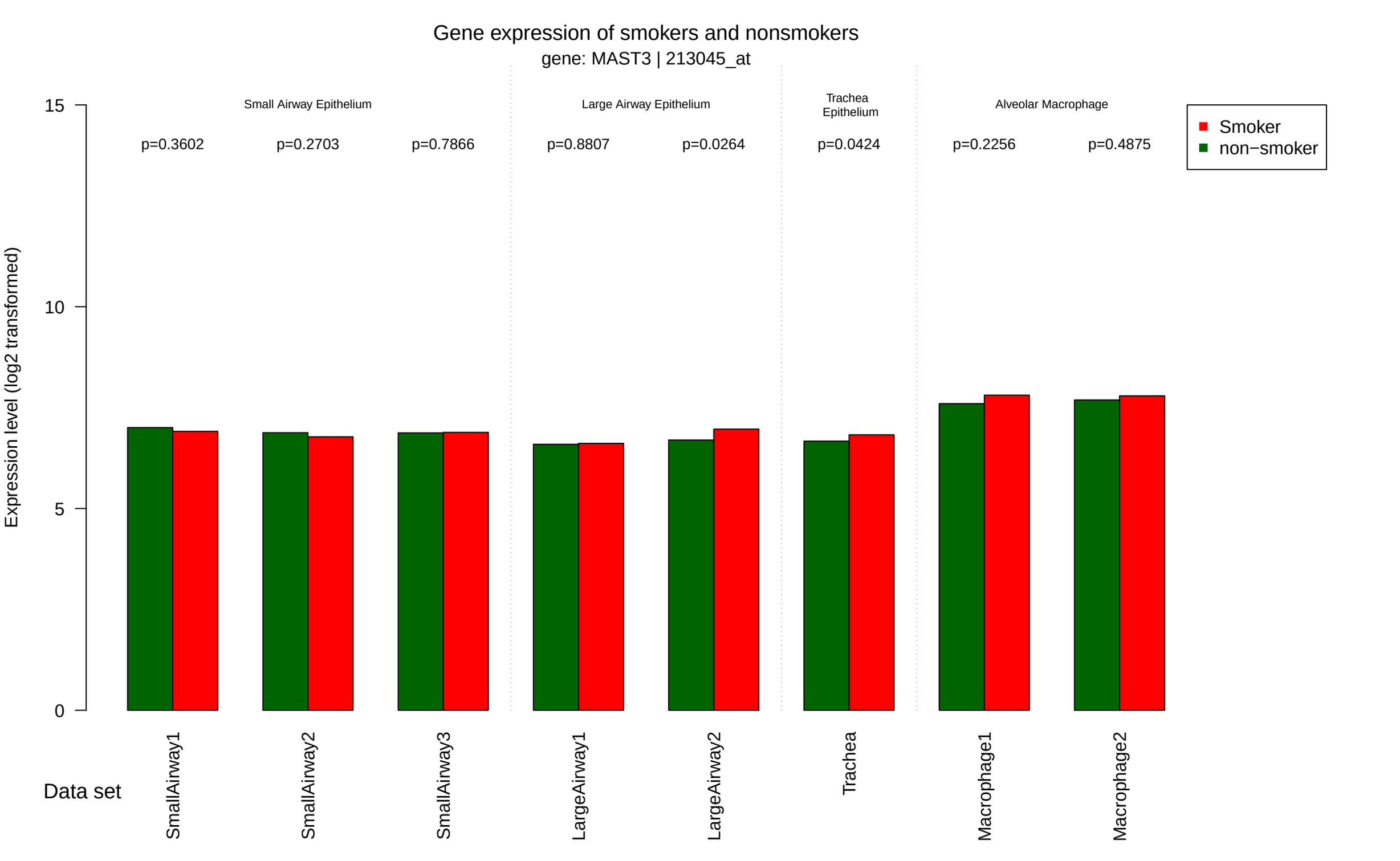
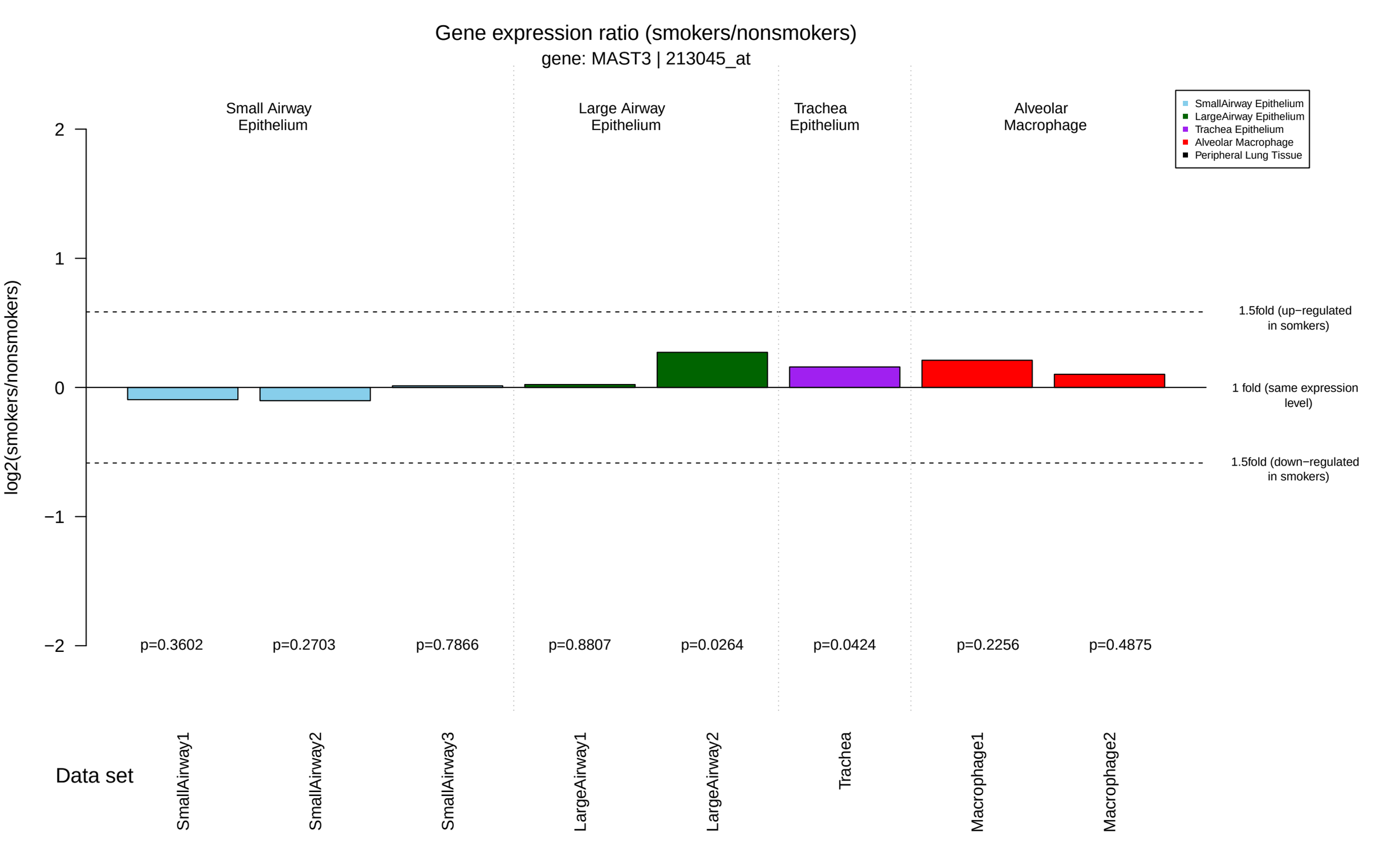
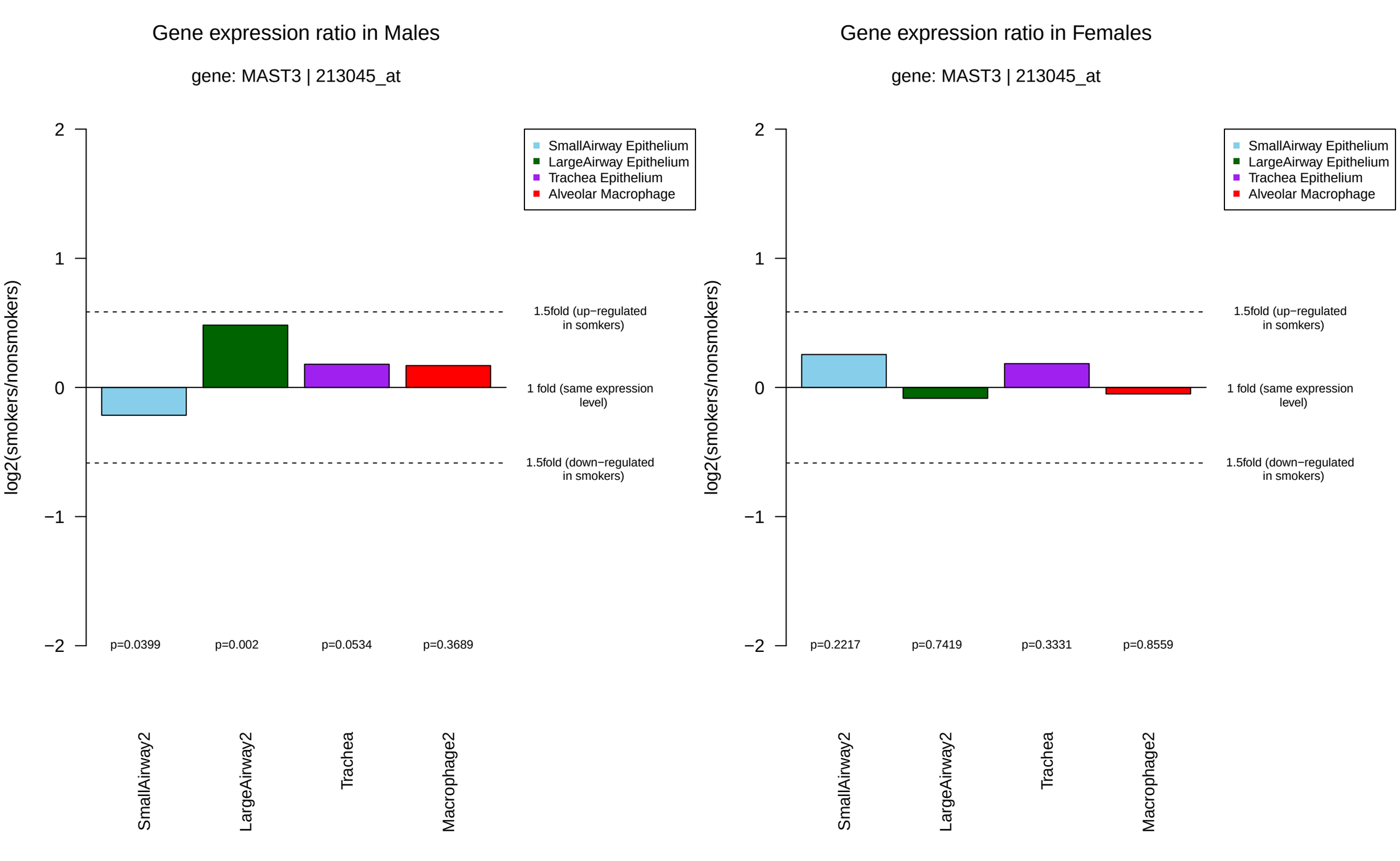
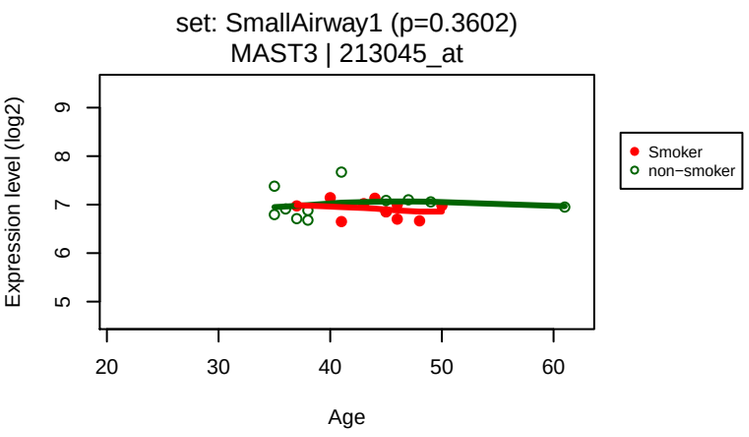
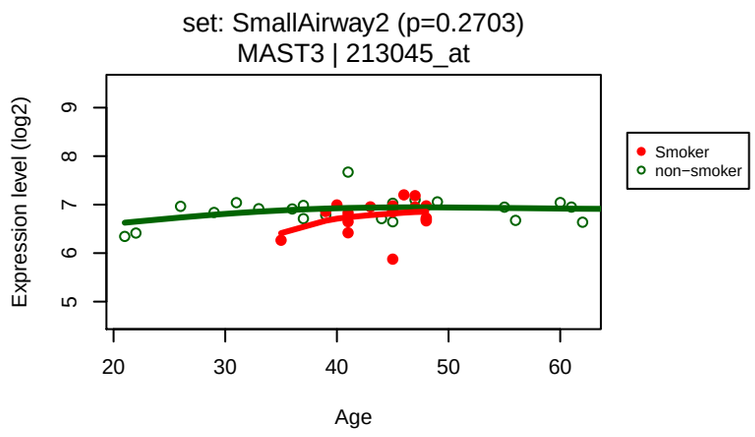
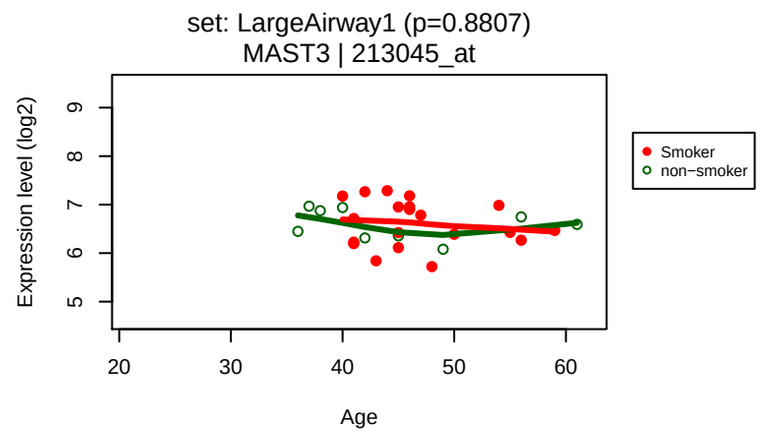

Gene expression of smokers and nonsmokers
gene: MAST3 | 213045_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3602
p=0.2703
p=0.7866
p=0.8807
p=0.0264
p=0.0424
p=0.2256
p=0.4875
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

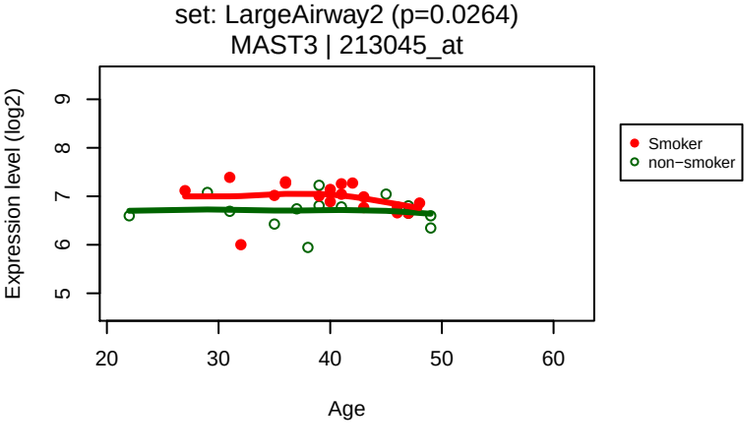

Gene expression ratio (smokers/nonsmokers)
gene: MAST3 | 213045_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3602
p=0.2703
p=0.7866
p=0.8807
p=0.0264
p=0.0424
p=0.2256
p=0.4875
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

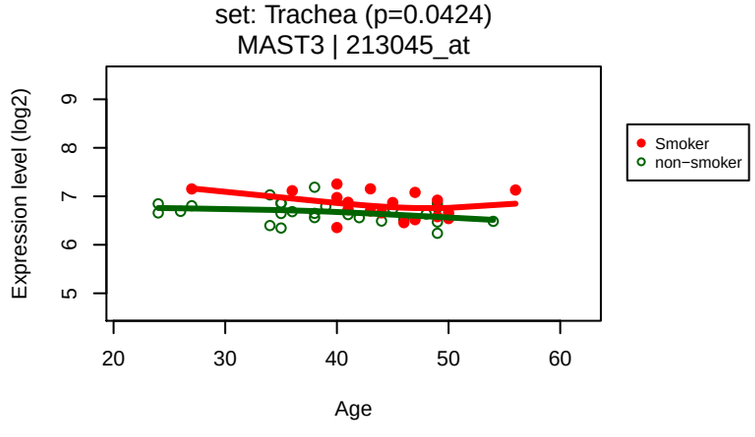

Gene expression ratio in Males
gene: MAST3 | 213045_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0399
p=0.002
p=0.0534
p=0.3689
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MAST3 | 213045_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2217
p=0.7419
p=0.3331
p=0.8559
SmallAirway2
LargeAirway2
Trachea
Macrophage2

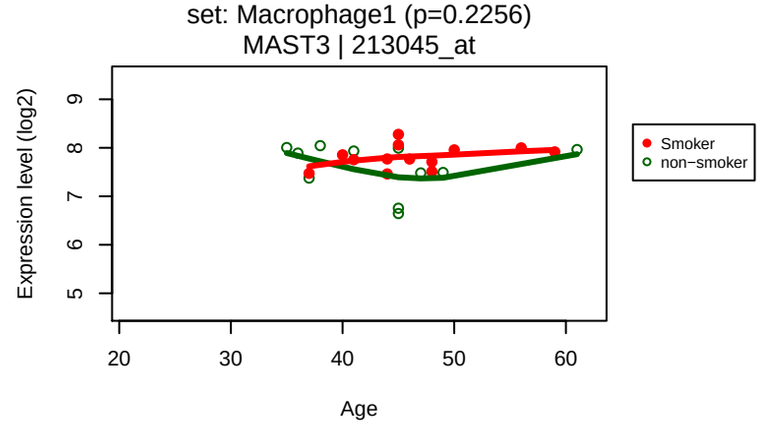

set: SmallAirway1 (p=0.3602)
MAST3 | 213045_at
Expression level (log2)
Age
Smoker
non−smoker

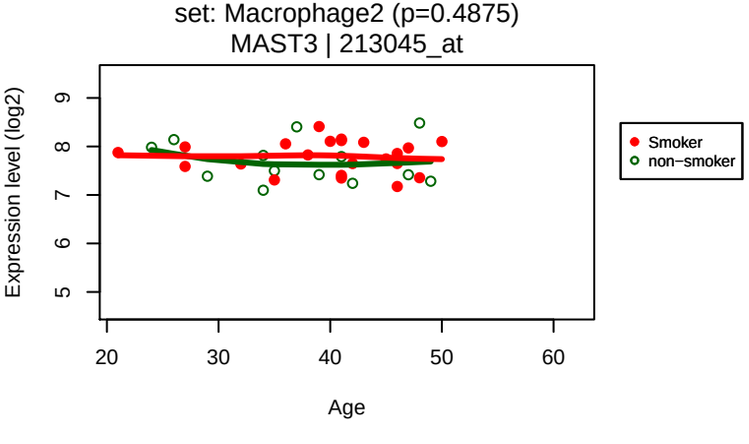

set: SmallAirway2 (p=0.2703)
MAST3 | 213045_at
Expression level (log2)
Age
Smoker
non−smoker

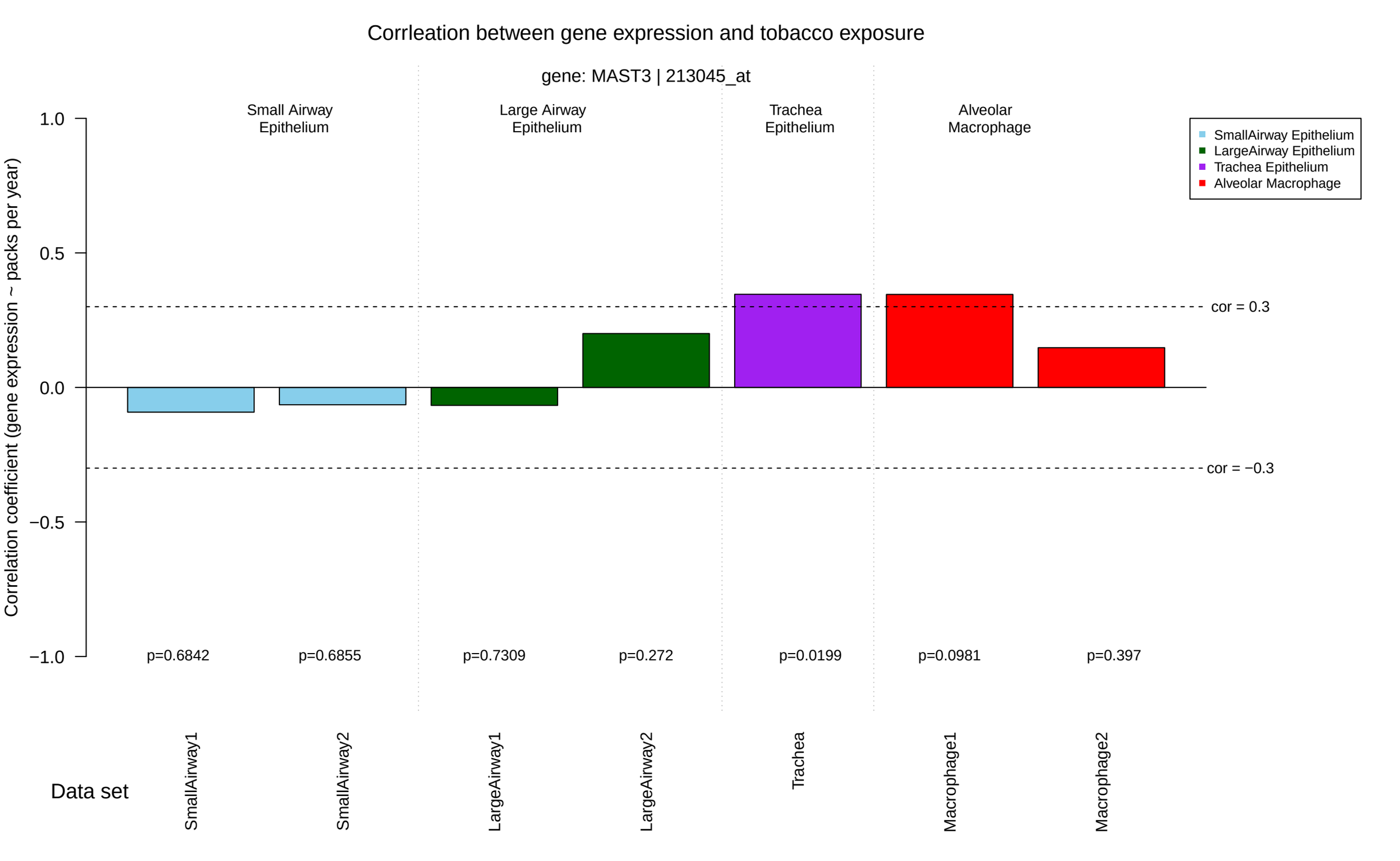

set: LargeAirway1 (p=0.8807)
MAST3 | 213045_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0264)
MAST3 | 213045_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0424)
MAST3 | 213045_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2256)
MAST3 | 213045_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4875)
MAST3 | 213045_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MAST3 | 213045_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6842
p=0.6855
p=0.7309
p=0.272
p=0.0199
p=0.0981
p=0.397
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2